Technical Document #71: Retrieval Augmented Generation - Comprehensive Overview

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Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.
- Semantic search finds documents based on meaning rather than exact keywords.
- Re-ranking models score retrieved documents for relevance to the query.